



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Team ID	EN15			
Semester	6 th			
Course :	Interdisciplinary Project			
Team Details		Program	USN	Name
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Project Title	Energy-Aware Digital Twin of LEO Satellite Constellations Using Scalable N-Body Simulation			
Center of Excellence				
Internal Guide				
Name Designation & Department	Dr. Rachana S Akki Associate Professor EEE			

INTRODUCTION:

Low Earth Orbit (LEO) satellite constellations are rapidly expanding with thousands of satellites deployed for global communication, navigation, and Earth observation. Managing these large constellations requires accurate simulation of orbital dynamics, satellite energy availability, and network reliability. However, most existing models analyse these aspects independently and lack a unified framework capable of simulating large-scale constellations in real time.

This project proposes the development of a digital twin of satellite constellations, where each satellite is modelled as an energy-aware computational node in a distributed ICT network. Real orbital data from Two-Line Element (TLE) sets will be propagated using the SGP4 algorithm, while a scalable N-body simulation engine will model orbital interactions and collision risks. The framework will also simulate cascading debris events (Kessler Syndrome) and evaluate constellation-level network availability and resilience.

OBJECTIVES:

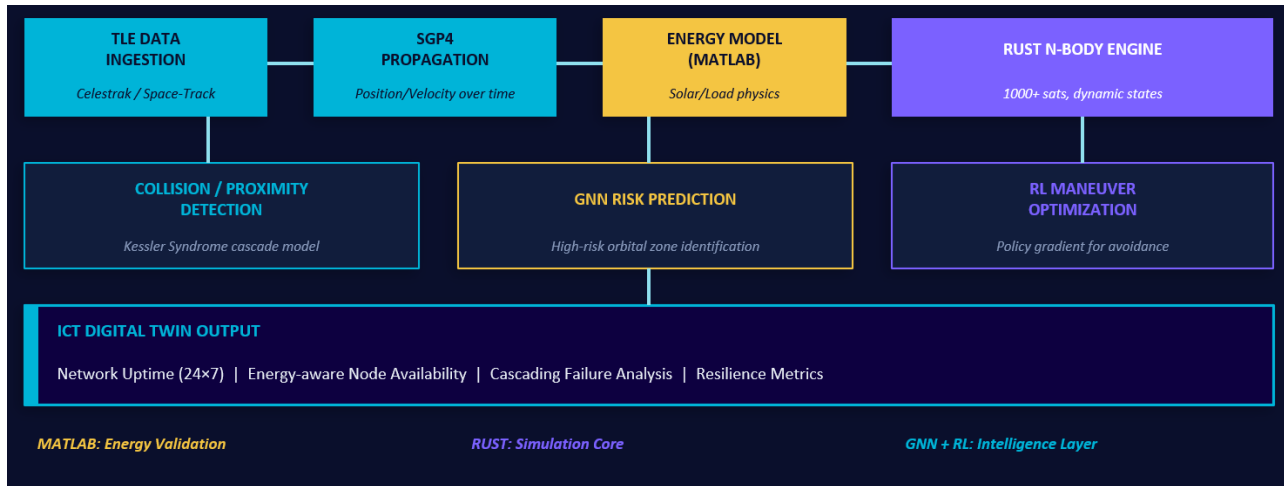
- To develop a scalable digital twin framework for Low Earth Orbit (LEO) satellite constellations using real TLE data and N-body orbital simulation.
- To analyse collision risks and network resilience in large-scale satellite constellations by integrating energy-aware satellite modelling and proximity detection mechanisms.



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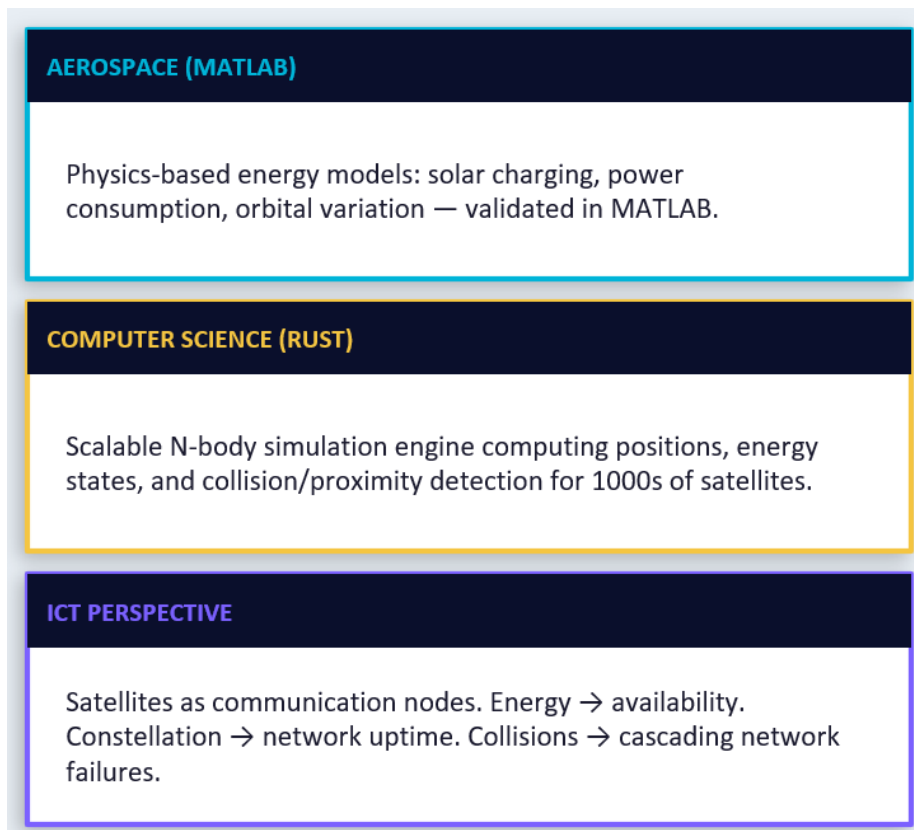
METHODOLOGY:



SOFTWARE REQUIREMENTS:

- MATLAB (Energy modelling and validation)
- Rust programming language (N-body simulation engine)
- Python (Machine learning and data analysis)
- PyTorch / TensorFlow (GNN and RL models)
- SGP4 orbital propagation library
- Git / GitHub for version control

INTERDISCIPLINARY RELEVANCE:



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INNOVATION / CONTRIBUTION TO THE FIELD:

The proposed work introduces a scalable digital twin framework for satellite constellations that simultaneously integrates orbital dynamics, energy modelling, network analysis, and AI-based decision-making.

Key contributions include:

- A Rust-based N-body simulation engine capable of modelling thousands of satellites.
- Integration of energy-aware satellite models with network availability analysis.
- Simulation of Kessler cascade events to study constellation resilience.
- AI-driven prediction of collision risks and optimization of avoidance maneuvers.

This framework can assist researchers and satellite operators in evaluating the safety and reliability of large-scale LEO constellations.

TIMELINE (GANTT CHART):

Week	Activity
W1–W2	Literature Survey & Problem Definition
W3–W4	TLE Data Processing & SGP4 Integration
W5–W6	MATLAB Energy Model Development
W7–W8	Development of N-Body Simulation Engine
W9–W10	Collision Detection & Debris Cascade Model
W11–W12	GNN Model Training for Risk Prediction
W13–W14	RL Maneuver Optimization
W15	Integration of Digital Twin Framework
W16	Testing, Documentation & Final Review



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Signature of:

Guide

HOD

**Dean Academics
Dr. M V Renukadevi**

Name:

Designation:

Department:

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